Testiddle

PRACTICE TEST

02

TT-SAT-02



SAT-STYLE PRACTICE SERIES

SAT *Practice Test*

Digital format, printed. Sit it in one go, on the clock, then mark it and learn from every miss.

4	120	164
MODULES	QUESTIONS	MINUTES

Before you start

- [ ] A sharp pencil, an eraser, and a watch or timer
- [ ] A calculator for the two Math modules
- [ ] A quiet table and about 184 minutes with no interruptions
- [ ] Circle your answers in this booklet, or use the answer sheet at the back

MODULE	QUESTIONS	TIME
Reading and Writing, Module 1	33	39 min
Reading and Writing, Module 2 (harder)	33	39 min
Math, Module 1 (calculator permitted)	27	43 min
Math, Module 2 (harder)	27	43 min

Every question in this test is original and written by Testiddle. It follows the common patterns of the real exam but is not affiliated with, or endorsed by, any exam provider.

Test begins on the next page.

Start your timer now.

1 Reading and Writing

33 QUESTIONS 39 MINUTES

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer. Circle your answer in the booklet.

PACE YOURSELF

You have **39 minutes** for **33 questions**, which is about **71 seconds** a question. Answer what you can quickly, circle the ones that need more time, and come back to them before time is called. There is no penalty for a wrong answer, so never leave a question blank.

WHAT YOU WILL SEE IN THIS MODULE

- Words in context
- Text structure and purpose
- Central ideas and details
- Command of evidence: quantitative
- Command of evidence: textual
- Inferences
- Form, structure, and sense
- Boundaries
- Transitions
- Rhetorical synthesis

Questions 1 to 4 of 33

1 WORDS IN CONTEXT

Seeking to _____ the spread of an invasive beetle, forestry officials restricted the movement of firewood between counties, since the insects often travel hidden inside cut logs.

Which choice completes the text with the most logical and precise word or phrase?

(A) encourage
(B) document
(C) limit
(D) celebrate

2 WORDS IN CONTEXT

The chef's menu is deliberately _____ : it lists only five dishes, each built from ingredients grown within a few miles of the restaurant.

Which choice completes the text with the most logical and precise word or phrase?

(A) limited
(B) sprawling
(C) indifferent
(D) outdated

3 WORDS IN CONTEXT

Though her early paintings were widely admired, the artist found that success had made her _____ ; she began to repeat herself, afraid of disappointing buyers who expected the same bright landscapes.

Which choice completes the text with the most logical and precise word or phrase?

(A) curious
(B) generous
(C) impatient
(D) complacent

4 WORDS IN CONTEXT

The oldest trees in the grove are so _____ that foresters have not found any seedlings growing within thirty metres of them; their dense shade prevents young trees from taking root.

Which choice completes the text with the most logical and precise word or phrase?

(A) delicate
(B) imposing
(C) scattered
(D) unremarkable

5 WORDS IN CONTEXT

Rather than _____ the old factory, the developers chose to convert it into apartments, keeping its brick walls and iron beams as features of the new design.

Which choice completes the text with the most logical and precise word or phrase?

(A) demolish

(B) inhabit

(C) photograph

(D) purchase

6 WORDS IN CONTEXT

Critics have called the senator's speech _____ : it was long and well delivered, but it contained no specific proposals and left listeners unsure what she actually intended to do.

Which choice completes the text with the most logical and precise word or phrase?

(A) precise

(B) vague

(C) provocative

(D) brief

7 TEXT STRUCTURE AND PURPOSE

The following text is adapted from an original science article.

> Honeybees communicate the location of food through a pattern of movement known as the waggle dance. The angle of the bee's run relative to vertical corresponds to the direction of the food relative to the sun, while the duration of the waggle signals the distance. Other bees in the hive follow the dancer, then fly out to search the indicated area.

Which choice best describes the function of the underlined sentence in the text as a whole?

(A) It questions whether bees can communicate with one another.

(B) It compares the dance with the way other insects find food.

(C) It explains how the features of the dance convey information about the food.

(D) It describes what happens when bees fail to find food.

8 TEXT STRUCTURE AND PURPOSE

The following text is from an original short story. Lucía has come to collect her late aunt's belongings.

> The flat smelled of lemon polish and old paper. Lucía moved from room to room with a box she did not fill, lifting a teacup, a scarf, a folded letter, and setting each one down again exactly where it had been. It seemed impossible to rearrange a life that had been so precisely arranged. In the kitchen, a calendar still showed the month of her aunt's last week, a dentist's appointment circled in red; Lucía found she could not turn the page.

Which choice best states the main purpose of the text?

A. To describe the carelessness with which Lucía treats her aunt's things

B. To explain why Lucía's aunt kept the flat so tidy

C. To suggest that Lucía is hurrying to finish an unpleasant task

D. To show Lucía's reluctance to disturb the traces of her aunt's life

9 TEXT STRUCTURE AND PURPOSE

The following text is from an original poem.

> The river keeps no calendar. It moves
> through winter and through harvest, unaware
> of boats or bridges, or the quiet grooves
> we cut along its banks and call a prayer.
> And yet we stand there, hoping it will stay
> a little while beside us, as though time
> were something one could ask to slow its way
> and, asked politely, answer with a chime.

Which choice best describes the overall structure of the text?

A. The speaker describes a river's history, then predicts how it will change.

B. The speaker presents the river's indifference to people, then reflects on a human hope that it might be otherwise.

C. The speaker complains about people's habits, then proposes a remedy.

D. The speaker contrasts two rivers, then explains which one is preferable.

10 CENTRAL IDEAS AND DETAILS

The following text is adapted from an original essay about urban gardening.

> On rooftops across the city, residents have begun growing tomatoes, beans and herbs in raised beds. The harvests are modest and will never feed a neighbourhood. But those who garden say the real yield is something else: the conversations over the fence, the sharing of seeds, the sense of belonging to a place they might otherwise only pass through.

Which choice best states the main idea of the text?

(A) Rooftop gardens can supply enough food for an entire neighbourhood.

(B) Growing tomatoes is more difficult in a city than in the countryside.

(C) Rooftop gardens are valued by their keepers less for the food they produce than for the community they create.

(D) Most city residents have taken up gardening in recent years.

11 CENTRAL IDEAS AND DETAILS

The following text is from an original essay about a historical figure.

> The engineer Rosalind Teague is remembered, if at all, for a single failure: a viaduct that closed after only six years. Yet her notebooks show a designer far ahead of her time, one who calculated the effects of vibration decades before it became standard practice. The viaduct failed in part because the contractor replaced her specified steel with a cheaper alloy; her calculations, applied correctly, would have kept it standing.

Which choice best states the main idea of the text?

(A) Teague's notebooks were destroyed after the viaduct closed.

(B) Teague argued that vibration should never be considered in bridge design.

(C) Teague was the first engineer to use steel in a viaduct.

(D) Teague's reputation rests on a failure that was caused largely by decisions made against her design.

12 COMMAND OF EVIDENCE: QUANTITATIVE

Fruit Preference by Age Group

Age group	Prefer loose fruit	Prefer pre-packed
Under 35	41%	59%
35 to 54	55%	45%
55 and over	72%	28%

Researchers asked 400 shoppers whether they prefer to buy fruit loose or pre-packed. A writer wants to claim that older shoppers are more likely than younger shoppers to prefer loose fruit.

Which choice most effectively uses data from the table to support the writer's claim?

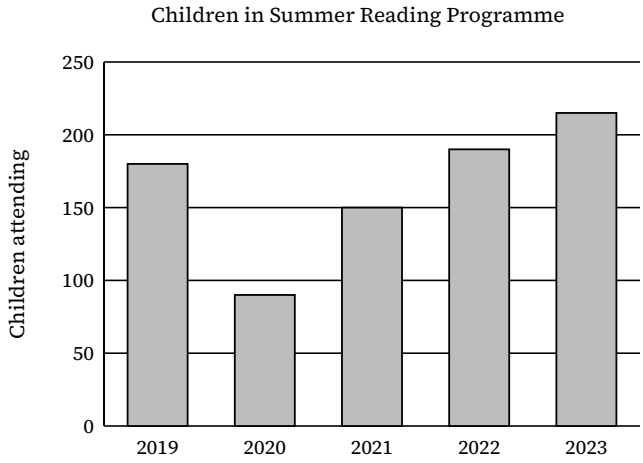
(A) Of shoppers aged 55 and over, 72 percent preferred loose fruit, compared with 41 percent of shoppers under 35.

(B) Of shoppers under 35, 59 percent preferred pre-packed fruit, compared with 28 percent of shoppers aged 55 and over.

(C) Of all the shoppers surveyed, 57 percent preferred loose fruit.

(D) Of shoppers aged 55 and over, 72 percent preferred loose fruit, and 41 percent of shoppers under 35 preferred pre-packed fruit.

13 COMMAND OF EVIDENCE: QUANTITATIVE



A town library tracked how many children attended its summer reading programme from 2019 to 2023. A librarian wants to emphasise the programme's recovery after a closure in 2020.

Which choice most effectively uses data from the graph to complete the text?

(A) attendance fell to 90 children in 2020 but rose to 215 by 2023, more than double its lowest point.

(B) attendance fell to 90 children in 2020 and was 215 in 2019.

(C) attendance rose from 90 children in 2020 to 150 in 2021, then stopped growing.

(D) attendance in 2023 was higher than in 2019, with 215 children taking part.

14 COMMAND OF EVIDENCE: TEXTUAL

The following text is adapted from an original essay about a composer.

> In her early symphonies, the composer Anneli Vos filled every bar with sound, layering strings over brass over percussion. Her last works could hardly be more different. One of them asks the orchestra to play almost nothing for nearly a minute, leaving the audience with only the faint creak of seats.

A student claims that Vos's later music is much sparser than her earlier music. Which quotation from the text best supports the student's claim?

(A) "filled every bar with sound, layering strings over brass over percussion"

(B) "the composer Anneli Vos"

(C) "leaving the audience with only the faint creak of seats"

(D) "One of them asks the orchestra to play almost nothing for nearly a minute"

15 COMMAND OF EVIDENCE: TEXTUAL

A student is writing about the discovery of a medieval manuscript.

> In 2018, a cataloguer at a regional archive found that a bundle of accounts, long assumed to be a farm's ledger, contained a fragment of a twelfth-century poem on its back pages. The fragment had been used as scrap to bind the ledger, a common practice at the time.

Which finding, if true, would most directly support the claim that the fragment was reused as binding material rather than copied into the ledger?

(A) The poem is written in a language no longer spoken.

(B) The ledger lists the sale of livestock.

(C) Its pages were folded and stitched around the ledger's cover.

(D) The archive holds other twelfth-century manuscripts.

16 COMMAND OF EVIDENCE: TEXTUAL

The following text is adapted from an original essay about tidal power.

> Proponents of tidal power point out that tides, unlike wind or sunshine, can be predicted centuries in advance, which makes the electricity they generate easy to plan around. Critics reply that the capital cost of a tidal barrage is enormous and that its effect on estuary wildlife is poorly understood.

Which choice best states a claim made by proponents of tidal power in the text?

(A) A tidal barrage is cheaper to build than a wind farm.

(B) Electricity from tides is more predictable than electricity from wind or sunshine.

(C) Estuary wildlife is unaffected by tidal power.

(D) Tides are stronger than winds.

17 INFERENCES

The following text is adapted from an original science article.

> Seeds of the fire poppy can lie dormant in soil for decades without sprouting. Researchers studying burned hillsides noticed that the flowers appeared in large numbers only after wildfires, and they found that chemicals in smoke trigger germination. This suggests that fire poppies

Which choice most logically completes the text?

(A) are adapted to take advantage of conditions created by fire.

(B) are unable to grow in any soil that has not burned.

(C) sprout every year regardless of the conditions.

(D) produce seeds that are harmed by smoke.

18 INFERENCES

The following text is from an original essay about reading habits.

> A survey of university students found that those who read for pleasure at least an hour a week scored higher on a test of vocabulary than those who did not. The researcher cautions, however, that students with strong vocabularies may simply find reading more enjoyable and so do more of it. The survey's results therefore

Which choice most logically completes the text?

(A) prove that reading for pleasure harms vocabulary.

(B) do not show by themselves that reading for pleasure improves vocabulary.

(C) show that students with weak vocabularies never read.

(D) apply only to students at one university.

19 INFERENCES

The following text is adapted from an original essay about a pianist.

> The pianist Tomás Rueda gave his first public recital at the age of eight and had performed in twenty countries by his twentieth birthday. Yet he has said that the most formative period of his training was a year in which he did not perform at all, studying scores in a rented room. It seems likely that Rueda believes that

Which choice most logically completes the text?

(A) young pianists should not give public recitals.

(B) travelling to other countries is harmful to a musician.

(C) scores are of little use to anyone who can already play well.

(D) deep study of the music matters more to a pianist's growth than frequent performance.

20 FORM, STRUCTURE, AND SENSE

The students in Ms. Alvarez's class _____ their projects on local history to the school board next week.

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) presents

(B) presented

(C) will present

(D) has presented

Questions 21 to 24 of 33

21 BOUNDARIES

The museum's collection includes paintings, sculptures, and _____ from every continent.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) textile that is
B) textiles which are
C) textiles
D) a textile

22 FORM, STRUCTURE, AND SENSE

Having studied the migration of monarch butterflies for a decade, _____

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Dr. Kim presented her findings at the annual meeting.
B) her findings were presented by Dr. Kim at the annual meeting.
C) the annual meeting heard Dr. Kim's findings.
D) it was Dr. Kim who presented findings at the meeting.

23 BOUNDARIES

The new bridge, which took nearly four years to build _____ finally opened to traffic in March.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) ;
B) ,
C) :
D) .

24 FORM, STRUCTURE, AND SENSE

Each of the volunteers _____ a badge at the entrance.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) receive
B) are receiving
C) were receive
D) receives

25 BOUNDARIES

Maya studied hard for the exam; _____ she earned the highest score in her class.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) however,
B) for example,
C) meanwhile,
D) as a result,

26 FORM, STRUCTURE, AND SENSE

The scientist's discovery was important _____ it changed how researchers thought about the origins of the disease.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) although
B) because
C) unless
D) whereas

27 BOUNDARIES

Among the many gifts the explorer received from the village _____ a carved wooden mask and a woven cloak.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) was
B) is
C) were
D) being

28 TRANSITIONS

Tomatoes are usually classed as vegetables because they are served with savoury food. Botanically, _____ they are fruit, since they develop from the flower's ovary and contain seeds.

Which choice completes the text with the most logical transition?

A) however,
B) for example,
C) likewise,
D) therefore,

29 TRANSITIONS

The city's new bike lanes were expected to reduce traffic. _____ commuting times have fallen by an average of five minutes since they opened.

Which choice completes the text with the most logical transition?

A) Indeed,

B) Otherwise,

C) Nevertheless,

D) Previously,

30 TRANSITIONS

Biologists once thought that all bats find their way by echolocation. _____ some fruit bats, which rely mainly on sight and smell, do not use it at all.

Which choice completes the text with the most logical transition?

A) Likewise,

B) In fact,

C) Instead,

D) Overall,

31 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- The Merrow Observatory stands on a hill above the town of Kell.
- It was built in 1912.
- Its main instrument is a 40-centimetre refracting telescope.
- The telescope is used to track variable stars.
- The observatory is open to the public on clear Friday evenings.

The student wants to explain what a particular building is for.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The Merrow Observatory, built in 1912, stands on a hill above the town of Kell.

B) The Merrow Observatory is open to the public on clear Friday evenings.

C) A telescope was installed at the observatory when it was built in 1912.

D) The Merrow Observatory above Kell uses a 40-centimetre refracting telescope to track variable stars.

32 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- The Orla River is 480 kilometres long.
- The Tamsin River is 310 kilometres long.
- The Orla flows through three countries.
- The Tamsin flows through one country.
- Both rivers empty into the same sea.

The student wants to make a comparison between two rivers.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The Orla River, which is 480 kilometres long, flows through three countries.

(B) Both the Orla and the Tamsin rivers empty into the same sea.

(C) The Orla River is 480 kilometres long and passes through three countries, while the Tamsin is 310 kilometres long and stays within one.

(D) The Tamsin River is 310 kilometres long and flows through only one country.

33 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- The Tarn Codex is held in a library in Lisbon.
- It is written on parchment.
- Scholars date it to around the year 1190.
- It contains 120 pages of astronomical tables.
- It was restored in 1985.

The student wants to emphasise the age of a manuscript.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Scholars date the Tarn Codex, which is held in a library in Lisbon, to around the year 1190.

(B) The Tarn Codex, restored in 1985, contains 120 pages of astronomical tables.

(C) The Tarn Codex is written on parchment and held in a library in Lisbon.

(D) The Tarn Codex contains astronomical tables on 120 pages of parchment.

STOP If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

1 Reading and Writing Module 2

33 QUESTIONS 39 MINUTES

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer. Circle your answer in the booklet.

PACE YOURSELF

You have **39 minutes** for **33 questions**, which is about **71 seconds** a question. Answer what you can quickly, circle the ones that need more time, and come back to them before time is called. There is no penalty for a wrong answer, so never leave a question blank. This module is harder than Module 1.

WHAT YOU WILL SEE IN THIS MODULE

- Words in context
- Text structure and purpose
- Cross-text connections
- Central ideas and details
- Command of evidence: quantitative
- Command of evidence: textual
- Inferences
- Form, structure, and sense
- Boundaries
- Transitions
- Rhetorical synthesis

Questions 1 to 4 of 33

1 WORDS IN CONTEXT

The ambassador's remarks were carefully _____ : she affirmed her country's friendship with its neighbour while avoiding any statement that might later be read as a commitment.

Which choice completes the text with the most logical and precise word or phrase?

(A) calibrated
(B) impulsive
(C) defiant
(D) garbled

2 WORDS IN CONTEXT

Once regarded as a _____ of the tropics, the disease has been spreading northward as warming temperatures allow its mosquito carriers to survive in regions that were formerly too cold.

Which choice completes the text with the most logical and precise word or phrase?

(A) haven
(B) remedy
(C) relic
(D) scourge

3 WORDS IN CONTEXT

The treaty's language was so _____ that each side could claim it had won a concession, a flexibility that smoothed the signing but sowed disputes for years afterward.

Which choice completes the text with the most logical and precise word or phrase?

(A) rigid
(B) elastic
(C) exhaustive
(D) emphatic

4 WORDS IN CONTEXT

Far from being _____ to criticism, the physicist actively sought it, circulating drafts of her papers to rivals and asking them to find flaws before publication.

Which choice completes the text with the most logical and precise word or phrase?

(A) vulnerable
(B) sensitive
(C) impervious
(D) indebted

5 WORDS IN CONTEXT

Although the physician's methods seemed _____ to her colleagues, who preferred established procedures, her patients' unusually quick recoveries gradually persuaded the hospital to adopt them.

Which choice completes the text with the most logical and precise word or phrase?

(A) conventional

(B) unorthodox

(C) negligent

(D) unnecessary

6 WORDS IN CONTEXT

Although the explorer's journals were long _____ as exaggerated, a recent survey of the coast he described found nearly every landmark exactly where he had placed it.

Which choice completes the text with the most logical and precise word or phrase?

(A) dismissed

(B) verified

(C) transcribed

(D) celebrated

7 TEXT STRUCTURE AND PURPOSE

The following text is adapted from an original essay about the history of science.

> We tend to imagine that a scientific revolution arrives with a single experiment, a decisive moment. But the shift in astronomy that replaced the earth-centred view of the cosmos unfolded over more than a century. Observers had to accumulate small discrepancies, one by one, until the old model required so many patches that its defenders themselves began to doubt it. Only then did the new view appear not as heresy but as relief.

Which choice best describes the function of the underlined sentence in the text as a whole?

(A) It describes a single experiment that settled a long-running dispute.

(B) It argues that the old model was never defended by astronomers.

(C) It explains how the slow accumulation of evidence, rather than one decisive moment, brought about the change.

(D) It introduces a new theory that the passage goes on to evaluate.

8

TEXT STRUCTURE AND PURPOSE

The following text is from an original novel. Corliss is a retired ferry captain who has been invited to the christening of a new vessel.

> Corliss stood at the back of the crowd, his good coat buttoned against a wind he could have told them about. The speeches praised the vessel's engines, its modern radar, its capacity for four hundred passengers. Nobody mentioned the harbour mouth, where the current doubled back on itself after the turn of the tide, or the shoal that had taken three boats in his time. He had the odd impression of attending the christening of a ship by people who had never been to sea.

Which choice best describes the function of the sentence beginning "Nobody mentioned the harbour mouth" in the text as a whole?

(A) It shows that Corliss has been asked to give a speech.

(B) It reveals that the vessel is more dangerous than earlier ones.

(C) It contrasts the harbour's present safety with its past dangers.

(D) It illustrates what Corliss feels the speakers have overlooked.

9

TEXT STRUCTURE AND PURPOSE

The following text is from an original poem.

> They told me language was a house, and I
> should furnish it with care, and keep it clean,
> and never let a stranger's accent lie
> across the threshold. Years went by. I've seen
> the house grow loud with children's borrowed words,
> with market slang, with jokes my mother made,
> with all the strangers' tongues like flocks of birds
> that nest in eaves and do not ask to stay.

Which choice best describes the overall structure of the text?

(A) The speaker praises a house, then explains how it was built.

(B) The speaker questions a friend's advice, then follows it.

(C) The speaker reports an instruction about language, then describes how experience has contradicted it.

(D) The speaker lists the languages she knows, then ranks them.

10 CROSS-TEXT CONNECTIONS

TEXT 1

Some art historians argue that the invention of photography freed painters from the duty to depict the world accurately. With the camera available to record a likeness, they say, painters were able to explore colour and form for their own sake, a development that led to the movements of the late nineteenth century.

TEXT 2

A historian of technology notes that the new painting styles were already emerging before photography became widespread. She argues that the camera did not cause painters to abandon realism so much as make visible an interest in perception that many of them already had.

Based on the texts, how would the historian in Text 2 most likely respond to the claim in Text 1?

(A) By agreeing that painters lost interest in realism only after the camera appeared

(B) By arguing that photography had no influence on painters at all

(C) By claiming that painters' new styles were invented by photographers

(D) By suggesting that the claim wrongly treats photography as the cause of changes that were already under way

11 CENTRAL IDEAS AND DETAILS

The following text is adapted from an original science article.

> The common view of cells as bags of chemicals floating in fluid has given way to a more crowded picture. Using new microscopes, researchers have found that many cellular reactions occur not in open solution but within droplets that form and dissolve like oil in water, concentrating the molecules that need to meet. These droplets lack membranes, yet they behave as distinct compartments, suggesting that cells organise their chemistry with far less physical structure than had been believed necessary.

Which choice best states the main idea of the text?

(A) Cells are less crowded than researchers once believed.

(B) Recent observations indicate that cells can organise chemical reactions in compartments that have no membranes.

(C) Droplets in cells form only when oil is present.

(D) New microscopes have made membranes unnecessary for cells.

12

CENTRAL IDEAS AND DETAILS

The following text is adapted from an original essay about economics.

> It is tempting to treat a country's gross domestic product as a measure of well-being, but the figure counts any spending as a gain. Money spent cleaning up an oil spill raises GDP; so does money spent on medicine for an illness caused by pollution. An economist who reads the figure as a measure of welfare must therefore explain why a society that has to spend more to repair damage should count itself better off.

Which choice best states the main idea of the text?

A. GDP is a poor measure of well-being because it counts spending that merely repairs harm as a gain.

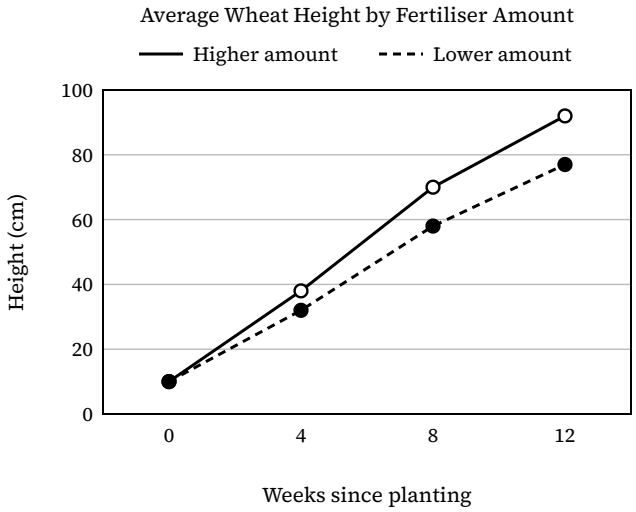
B. GDP falls whenever a society spends money on repairs.

C. Economists have abandoned GDP as a measure of national income.

D. Spending on medicine is the main source of economic growth.

13

COMMAND OF EVIDENCE: QUANTITATIVE



A group of agronomists measured the average height of wheat plants grown with two different amounts of nitrogen fertiliser over 12 weeks.

Which choice most effectively uses data from the graph to complete the statement?

A. By week 12, plants given the higher amount had grown to 92 centimetres, about 15 centimetres taller than plants given the lower amount.

B. By week 12, plants given the higher amount had grown to 92 centimetres, about 30 centimetres taller than plants given the lower amount.

C. By week 12, plants given the lower amount had grown to 92 centimetres, about 15 centimetres taller than plants given the higher amount.

D. By week 8, plants given the higher amount had grown to 92 centimetres, about 15 centimetres taller than plants given the lower amount.

14 COMMAND OF EVIDENCE: QUANTITATIVE

Average Journey Time (minutes) by Mode of Transport

Mode	Rush hour	Other times
Tram	31	29
Bus	44	30
Car	33	24

A transport researcher compared the average journey time of three ways of crossing a city during rush hour and non-rush hours. She wants to argue that the tram is the most reliable option.

Which choice most effectively uses data from the table to support the researcher's claim?

(A) The tram's average journey time of 31 minutes in rush hour was shorter than the bus's 44 minutes.

(B) The tram's average journey time differed by 2 minutes between rush hour and other times, while the car's differed by 14 minutes.

(C) The tram's average journey time differed by only 2 minutes between rush hour and other times, compared with 14 minutes for the bus and 9 minutes for the car.

(D) The car's average journey time of 24 minutes outside rush hour was the shortest of the three.

15 COMMAND OF EVIDENCE: TEXTUAL

The following text is adapted from an original essay on the history of medicine.

> Before germ theory, physicians commonly attributed fevers to bad air, and their remedies were aimed at purifying it: burning herbs, opening windows, relocating patients to hills. These measures sometimes worked, though not for the reason physicians believed; moving a sick person out of a crowded ward also reduced the chance of spreading infection to others.

A student claims that remedies based on a mistaken theory could still be effective. Which quotation from the text best supports the student's claim?

(A) "Before germ theory, physicians commonly attributed fevers to bad air"

(B) "burning herbs, opening windows, relocating patients to hills"

(C) "moving a sick person out of a crowded ward also reduced the chance of spreading infection"

(D) "These measures sometimes worked, though not for the reason physicians believed"

16 COMMAND OF EVIDENCE: TEXTUAL

A researcher claims that urban birds sing at a higher pitch than rural birds of the same species because lower notes are masked by traffic noise.

> In a study of great tits, the average song pitch was found to be 12 percent higher in a noisy city than in a quiet forest. When researchers played recordings of traffic to forest birds, the birds raised their pitch within a few days.

Which finding from the text most directly supports the researcher's claim about the cause of the higher pitch?

A. City birds sing more often than forest birds.

B. Forest birds raised their pitch after being exposed to recordings of traffic.

C. The study was carried out on great tits.

D. Great tits are common in both cities and forests.

17 COMMAND OF EVIDENCE: TEXTUAL

The following text is from an original essay about memory.

> People often feel that recalling a memory is like playing back a recording. Experiments suggest otherwise. When participants were asked to recall a film clip and were then given false details by an interviewer, many later reported the false details as part of what they had seen. Memory, the psychologist concluded, is rebuilt at each recall and is therefore open to revision.

Which choice best describes what the experiment described in the text showed?

A. That a memory can be changed by information supplied after the event

B. That participants forget film clips within minutes

C. That interviewers have better memories than participants

D. That recalling a memory exactly is easy with practice

18 INFERENCES

The following text is adapted from an original essay about urban history.

> Many medieval towns were built within defensive walls, and their streets are often narrow and winding. A historian notes that straighter, wider streets appeared in towns that expanded outside the walls once the threat of attack had lessened. This pattern suggests that the layout of the earlier streets

Which choice most logically completes the text?

(A) was designed to make travel across the town faster.

(B) was imposed by the same architects who built the later streets.

(C) was shaped, at least in part, by the need for defence.

(D) had little to do with the position of the walls.

19 INFERENCES

The following text is adapted from an original science article.

> A team studying a coastal marsh found that where cordgrass grew densely, the marsh surface rose by several millimetres a year, keeping pace with sea level. Where grass cover was thin, the surface barely rose, and the marsh was gradually drowned. The researchers conclude that restoring cordgrass is likely to

Which choice most logically completes the text?

(A) cause marshes to sink more quickly in the future.

(B) help marshes survive sea-level rise by raising their surface.

(C) make the marsh surface lower in areas of dense grass.

(D) be unnecessary wherever the sea is calm.

20 INFERENCES

The following text is adapted from an original essay about the invention of the printing press.

> Before the press, a scribe could produce only a few books a year, and the text of any work drifted slightly with each copying. After the press, thousands of identical copies of a single edition could circulate at once. Scholars could now cite a page number and be confident that a reader far away would find the same words on the same page. This suggests that the press made it possible for scholars to

Which choice most logically completes the text?

(A) avoid reading the texts of earlier scholars.

(B) write works that no reader could verify.

(C) produce handwritten copies faster than before.

(D) refer precisely to a shared text when discussing and checking one another's claims.

21 FORM, STRUCTURE, AND SENSE

The researchers, _____ had spent six months at the remote station, returned with samples from more than two hundred sites.

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) whom

(B) who

(C) which

(D) that they

22 BOUNDARIES

The number of species recorded in the park _____ risen steadily since the protection programme began.

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) have

(B) are

(C) has

(D) were

23 FORM, STRUCTURE, AND SENSE

Unlike most of her contemporaries, who wrote about city life, _____

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) Ottoline Pryce wrote almost exclusively about the countryside.

(B) the countryside was the almost exclusive subject of Ottoline Pryce.

(C) the writing of Ottoline Pryce was almost exclusively the countryside.

(D) Ottoline Pryce's subject, almost exclusively the countryside, was written about.

24 BOUNDARIES

The exhibition features work by three photographers: Aiko Mori, whose subjects are urban; Daniel Okoro, _____ Ruth Weill, who photographs only the sea.

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) whose portraits are staged, and

(B) whose portraits are staged and

(C) whose portraits are staged; with

(D) whose portraits are staged; and

25 BOUNDARIES

A tiny fraction of the sunlight that reaches the forest floor _____ enough to power the growth of the shade-tolerant ferns.

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) are

(B) is

(C) being

(D) have been

26 FORM, STRUCTURE, AND SENSE

Rarely _____ a comet bright enough to be seen without a telescope.

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) an observer encounters

(B) has an observer encountering

(C) do an observer encounter

(D) does an observer encounter

27 FORM, STRUCTURE, AND SENSE

The council has announced that it will begin repairing _____ streets next year, starting with those most damaged by floods.

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) the citys

(B) the cities

(C) the city's

(D) the cities'

28 TRANSITIONS

Nutritionists have long recommended whole grains over refined grains. _____ some recent studies suggest that the benefits are smaller than once believed, because people who choose whole grains also tend to exercise more and smoke less.

Which choice completes the text with the most logical transition?

(A) More recently,

(B) For this reason,

(C) In other words,

(D) Likewise,

29 TRANSITIONS

The early railways were treated by many as a threat to the countryside. _____ several poets and painters of the period embraced them, finding in the steam engine a new kind of beauty.

Which choice completes the text with the most logical transition?

(A) Nevertheless,

(B) Therefore,

(C) Similarly,

(D) Specifically,

30 TRANSITIONS

Scientists have found that the common octopus can solve puzzles, use tools and recognise individual humans. _____ its lifespan is only about two years, leaving little time for what it learns to be passed on.

Which choice completes the text with the most logical transition?

(A) Thus

(B) Moreover

(C) Yet

(D) Accordingly

31 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- The Nazca Lines are huge drawings etched into a desert plain in Peru.
- They were made between about 500 BCE and 500 CE.
- For decades, researchers could identify about 300 figures.
- In 2019, machine learning software helped find more than 140 previously unknown figures.
- The new figures are mostly small and were probably meant to be seen from the ground.

The student wants to highlight how a discovery changed an understanding of an ancient culture.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The Nazca Lines, which are huge drawings in Peru, were made between about 500 BCE and 500 CE.

(B) The new figures at Nazca are mostly small and were probably meant to be seen from the ground.

(C) Machine learning software was used in 2019 to study the Nazca Lines.

(D) The Nazca Lines, which for decades were thought to contain about 300 figures, are now known to include more than 140 additional ones identified in 2019 with machine learning software.

32 RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- A study tested a new method for detecting early heart disease.
- The method uses a 10-minute scan of the heart's electrical activity.
- In a sample of 2,000 adults, the method correctly identified 94 percent of those with disease.
- It also flagged 12 percent of healthy adults as having disease.
- The researchers say further trials are needed.

The student wants to present a finding to an audience familiar with the topic who want the key result and its limit.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) A study of 2,000 adults tested a 10-minute scan of the heart's electrical activity as a method of detecting early heart disease.

(B) In a study of 2,000 adults, a 10-minute scan correctly identified 94 percent of those with heart disease, though it also wrongly flagged 12 percent of healthy adults.

(C) The researchers who tested the scan say that further trials are needed.

(D) A 10-minute scan correctly identified 94 percent of those with disease in a sample of 2,000 adults.

33

RHETORICAL SYNTHESIS

While researching a topic, a student has taken the following notes:

- Some farmers plough their fields before planting.
- Others use no-till farming, planting seeds directly into the previous crop's stubble.
- Ploughing loosens the soil but exposes it to erosion.
- No-till farming leaves the soil structure intact and reduces erosion.
- No-till farming often requires more herbicide.

The student wants to contrast two approaches to a task.

Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Whereas ploughing loosens the soil but exposes it to erosion, no-till farming leaves the soil intact and reduces erosion, though it often requires more herbicide.

B) Some farmers plough their fields before planting, and others use no-till farming.

C) No-till farming reduces erosion and often requires more herbicide.

D) Ploughing loosens the soil, which is why some farmers do it before planting.

STOP If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

2 Math

27 QUESTIONS 43 MINUTES

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

NOTES

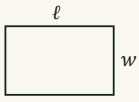
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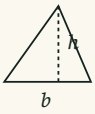
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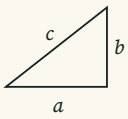
$A = \pi r^2$
$C = 2\pi r$



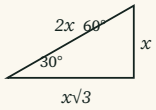
$A = \ell w$

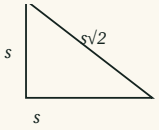


$A = \frac{1}{2}bh$

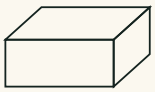


$c^2 = a^2 + b^2$

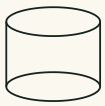




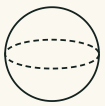
Special right triangles



$V = \ell wh$



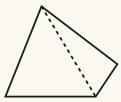
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360. The number of radians of arc in a circle is 2π. The sum of the measures in degrees of the angles of a triangle is 180.

ANSWERING THE QUESTIONS

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in the booklet. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer in the box under the question.

- If you find **more than one correct answer,** write only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long, write the decimal equivalent.
- If your answer is a **decimal** that is too long, truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as 3½), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Do not include **symbols** such as a percent sign, comma, or dollar sign in your answer.

PACE YOURSELF

You have **43 minutes** for **27 questions**, which is about **96 seconds** a question. Answer what you can quickly, circle the ones that need more time, and come back to them before time is called. There is no penalty for a wrong answer, so never leave a question blank.

WHAT YOU WILL SEE IN THIS MODULE

Percentages | Data analysis | Linear equations | Systems of equations | Functions | Nonlinear expressions | Ratios and rates | Triangles | Circles | Quadratics | Probability | Statistics | Scatterplots | Inequalities | Exponential functions | Linear functions | Percent change | Volume | Right triangles | Area

Questions 1 to 5 of 27

1 PERCENTAGES

What is 20% of 500?

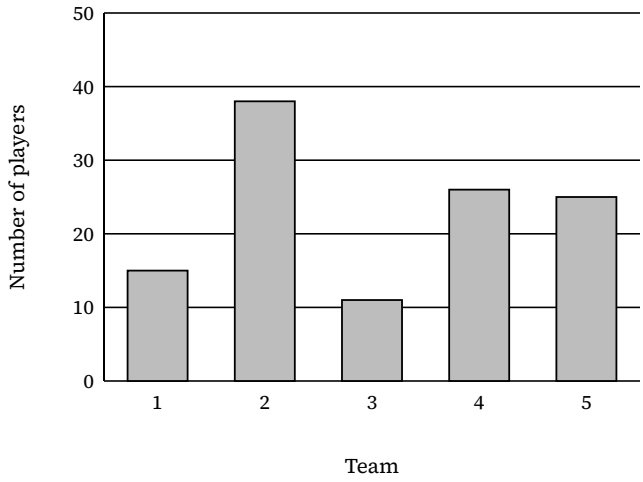
A) 400

B) 100

C) 200

D) 520

2 DATA ANALYSIS



The bar graph shows the number of players in each of five teams. How many more players does Team 5 have than Team 1?

A) 40

B) 20

C) 25

D) 10

3 LINEAR EQUATIONS

If $9x + 2 = 2x + 72$, what is the value of x?

A) 10

B) 12

C) 90

D) 20

4 SYSTEMS OF EQUATIONS

$x + y = 17$
$x - y = 11$

What is the value of x?

A) 3

B) 17

C) 14

D) 11

5 FUNCTIONS

x	1	2	3	4
f(x)	14	20	26	32

The table shows four values of the linear function f. Which equation defines f?

A) $f(x) = 8x + 6$

B) $f(x) = 6x + 14$

C) $f(x) = 6x + 8$

D) $f(x) = 14x$

6 NONLINEAR EXPRESSIONS

Which expression is equivalent to $(5x^5)(5x^6)$?

A $25x^{11}$

B $10x^{11}$

C $25x^{30}$

D $10x^{30}$

7 RATIOS AND RATES

A recipe uses 7 cups of flour for every 2 cups of sugar. How many cups of flour are needed for 6 cups of sugar?

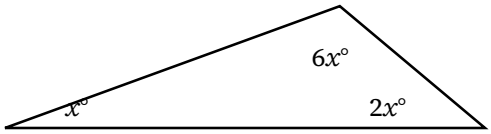
A 23

B 10

C 6

D 21

8 TRIANGLES



In the triangle shown, the three angles measure $x°$, $2x°$, $6x°$. What is the value of x?

A 15

B 20

C 30

D 60

9 CIRCLES

A circle has a circumference of 22π centimetres. What is the area of the circle, in square centimetres?

A 22π

B 11π

C 484π

D 121π

10 QUADRATICS

What are the solutions to $x^2 - 4x - 32 = 0$?

A) −4 and 8

B) 8 and 4

C) −8 and −4

D) −8 and 4

11 PROBABILITY

How Students Travel to School

	Walk	Bike	Bus
Grade 9	18	27	19
Grade 10	28	18	28

The table shows how 138 students travel to school. If one student who goes by bike is chosen at random, what is the probability that the student is in Grade 10?

A) $\frac{18}{138}$

B) $\frac{27}{45}$

C) $\frac{2}{5}$

D) $\frac{45}{138}$

12 STATISTICS

The mean of 5 test scores is 69. A score of 81 is added. What is the mean of all 6 scores?

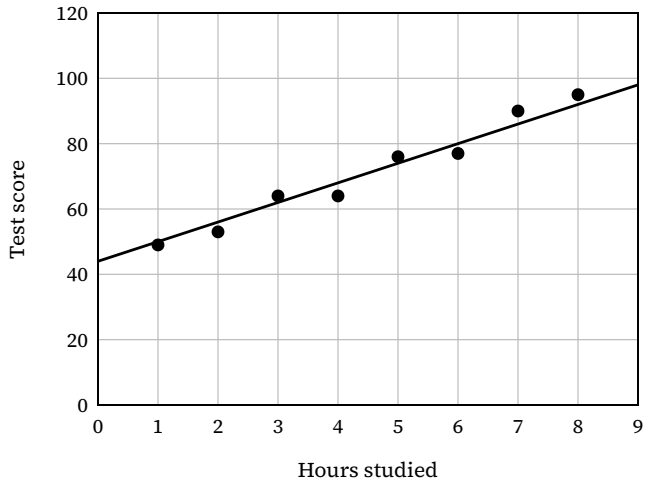
A) 75

B) 71

C) 70

D) 74

13 SCATTERPLOTS



The scatterplot shows the hours students studied and their test scores. The line of best fit is $y = 6x + 44$. According to the line, what is the predicted score for a student who studies 6 hours?

A) 36

B) 74

C) 80

D) 56

14 INEQUALITIES

A club needs at least 150 members to enter a competition, and 32 have already joined. If 10 new members join each week, which inequality represents the number of weeks, w, needed to reach the goal?

A) $32 + 10w \leq 150$

B) $32 + 10w \geq 150$

C) $10 + 32w \geq 150$

D) $150 + 10w \geq 32$

15 FUNCTIONS

The function f is defined by $f(x) = 4x^2 - 3x + 6$. What is the value of $f(4)$?

A) 82

B) 10

C) 64

D) 58

16 EXPONENTIAL FUNCTIONS

A population of bacteria triples every 6 hours. There are 800 bacteria at the start. Which function gives the number of bacteria after t hours?

A) $P(t) = 800(3)^{t/6}$

B) $P(t) = 800(3)^{6t}$

C) $P(t) = 800(6)^{t/3}$

D) $P(t) = 2400(t/6)$

17 LINEAR FUNCTIONS

A line in the xy-plane passes through the points (3, 13) and (6, 28). Which equation represents the line?

A) $y = 6x - 2$

B) $y = 5x$

C) $y = 5x - 2$

D) $y = 4x + 2$

18 PERCENT CHANGE

A town's population grew from 6,000 to 7,500. By what percent did the population increase?

A) 30%

B) 25%

C) 50%

D) 20%

19 VOLUME

A cylinder has a radius of 6 centimetres and a height of 8 centimetres. What is its volume, in cubic centimetres?

A) 288π

B) 96π

C) 48π

D) 864π

20 RIGHT TRIANGLES

A right triangle has legs of length 5 and 12. What is the length of its hypotenuse?

(A) 17

(B) 169

(C) 7

(D) 13

21 SYSTEMS OF EQUATIONS

The sum of two numbers is 22, and their difference is 10. What is the larger of the two numbers?

Write your answer

22 QUADRATICS

What is the positive solution to the equation $x^2 - x - 12 = 0$?

Write your answer

23 PERCENT CHANGE

The price of a concert ticket rose from \$20 to \$23. By what percent did the price increase?

Write your answer

24 LINEAR EQUATIONS

If $4(x - 2) = 8x - 52$, what is the value of x?

Write your answer

25 FUNCTIONS

The function f is defined by $f(x) = 8x - 6$. If $f(a) = 66$, what is the value of a?

Write your answer

26 RIGHT TRIANGLES

In a right triangle, one leg has length 6 and the hypotenuse has length 10. What is the length of the other leg?

Write your answer

27 AREA

What is the area, in square units, of a triangle with a base of 16 and a height of 7?

Write your answer

STOP If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

2 Math Module 2

27 QUESTIONS 43 MINUTES

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

NOTES

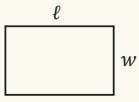
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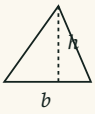
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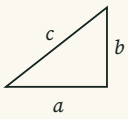
$A = \pi r^2$
$C = 2\pi r$



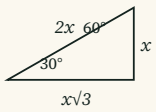
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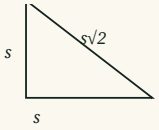


$A = \frac{1}{2}bh$

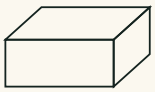


$c^2 = a^2 + b^2$

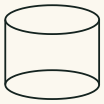




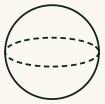
Special right triangles



$V = \ell wh$



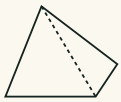
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360. The number of radians of arc in a circle is 2π. The sum of the measures in degrees of the angles of a triangle is 180.

Questions 1 to 6 of 27

1 EXPONENTIAL FUNCTIONS

A population of bacteria triples every 2 hours. There are 500 bacteria at the start. Which function gives the number of bacteria after t hours?

(A) $P(t) = 500(3)^{t/2}$

(B) $P(t) = 500(3)^{2t}$

(C) $P(t) = 500(2)^{t/3}$

(D) $P(t) = 1500(t/2)$

2 QUADRATICS

What are the solutions to $x^2 - 7x - 8 = 0$?

(A) 8 and 1

(B) −8 and −1

(C) −1 and 8

(D) −8 and 1

3 PERCENTAGES

What is 80% of 300?

(A) 60

(B) 480

(C) 380

(D) 240

4 VOLUME

A cylinder has a radius of 3 centimetres and a height of 7 centimetres. What is its volume, in cubic centimetres?

(A) 42π

(B) 63π

(C) 21π

(D) 189π

5 FUNCTIONS

The function f is defined by $f(x) = 3x^2 - 6x + 5$. What is the value of $f(4)$?

(A) 77

(B) −7

(C) 29

(D) 34

6 STATISTICS

The mean of 6 test scores is 80. A score of 94 is added. What is the mean of all 7 scores?

(A) 82

(B) 87

(C) 81

(D) 85

7 PROBABILITY

How Students Travel to School

	Walk	Bike	Bus
Grade 9	12	12	21
Grade 10	16	25	13

The table shows how 99 students travel to school. If one student who goes by bike is chosen at random, what is the probability that the student is in Grade 10?

A $\frac{25}{99}$

B $\frac{12}{37}$

C $\frac{37}{99}$

D $\frac{25}{37}$

8 PERCENTAGES

The price of an item is raised by 25%. Later, the new price is reduced by 40%. Compared with the original price, the final price is

A 15% lower

B 25% lower

C 65% lower

D the same as the original price

9 EQUIVALENT EXPRESSIONS

Which expression is equivalent to $(4x + 3)^2 - (3x - 4)^2$?

A $7x^2 - 7$

B $7x^2 + 48x + 25$

C $7x^2 + 48x - 7$

D $25x^2 + 48x - 7$

10 PERCENTAGES

150 is p% greater than 60. What is the value of p?

A 150

B 250

C 60

D 90

11 SYSTEMS OF EQUATIONS

$ax + 3y = 7$
$15x + 9y = 25$

In the given system of equations, a is a constant. If the system has no solution, what is the value of a?

A 15

B 8

C 9

D 5

12 SURFACE AREA

Two identical cubes are glued together along one full face to form a rectangular solid. The surface area of the solid is 640 square centimetres. What is the edge length, in centimetres, of each cube?

A) 10

B) 8

C) 7

D) 11

13 CIRCLES

A sector of a circle has a radius of 9 and a central angle of $\frac{\pi}{6}$ radians. What is the length of the arc of the sector?

A) $\frac{9\pi}{6}$

B) 9π

C) 13π

D) 18π

14 EQUIVALENT EXPRESSIONS

Which expression is equivalent to $\frac{x^2 - 12x + 35}{x - 7}$, where $x \neq 7$?

A) $x + 5$

B) $x - 5$

C) $x - 7$

D) $x - 35$

15 TRIGONOMETRY

In a right triangle, angle A is acute and $\sin A = \frac{3}{5}$. What is the value of $\tan A$?

A) $\frac{4}{5}$

B) $\frac{4}{3}$

C) $\frac{5}{3}$

D) $\frac{3}{4}$

16 EXPONENTIAL EQUATIONS

If $2^{x+4} = 4^{x-3}$, what is the value of x?

A) 7

B) 2

C) 10

D) 12

17 EXPONENTIAL FUNCTIONS

$V(x) = 12{,}000(0.9)^{x/6}$

The function V gives the value, in dollars, of a certain machine x months after it is purchased. By what percent does the value of the machine decrease each year?

A) 19%

B) 90%

C) 10%

D) 81%

18 LINEAR FUNCTIONS IN CONTEXT

$y = 3.5x + 45$

The equation gives the total cost y, in dollars, to rent a bicycle for x hours. Which statement is the best interpretation of 3.5 in this context?

A. The total cost of renting for 1 hour is $3.5.

B. The cost of each additional hour is $3.5.

C. The fixed fee for renting a bicycle is $3.5.

D. The cost of renting for 3.5 hours is $45.

19 ABSOLUTE VALUE

How many integers x satisfy the inequality $|x - 6| \leq 3$?

A. 6

B. 4

C. 3

D. 7

20 RATIOS AND RATES

One pump can fill a tank in 12 hours, and a second pump can fill it in 6 hours. Working together, how many hours do the two pumps take to fill the tank?

A. 9

B. 18

C. 4

D. 6

21 LINEAR EQUATIONS

If $4(x - 3) = 8x - 52$, what is the value of x?

Write your answer

22 CIRCLES IN THE PLANE

$x^2 + y^2 + 2x + 10y = 38$

The equation of a circle in the xy-plane is shown. What is the radius of the circle?

Write your answer

23 QUADRATICS

$2x^2 - 16x - 18 = 0$

What is the sum of the solutions to the given equation?

Write your answer

24 QUADRATICS

The height h, in metres, of a ball t seconds after it is thrown is given by $h(t) = -4t^2 + 48t + 10$. What is the maximum height, in metres, that the ball reaches?

Write your answer

Questions 25 to 27 of 27

25 PERCENTAGES

A solution that is 20% acid is mixed with a solution that is 70% acid to make a mixture that is 40% acid. If 10 litres of the 70% solution are used, how many litres of the 20% solution are needed?

Write your answer

26 QUADRATICS

The equation $x^2 + bx + 49 = 0$, where b is a positive constant, has exactly one real solution. What is the value of b?

Write your answer

27 SYSTEMS WITH NONLINEAR EQUATIONS

$y = x^2 + 2x + 6$
$y = 8x + k$

In the given system of equations, k is a constant. If the system has exactly one solution, what is the value of k?

Write your answer

STOP If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

Answer sheet

SAT Practice Test 2, Reading and Writing

Fill in the circle for each answer, or write your student-produced answers in the boxes. Then mark yourself with the answer key.

Module 1

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D

Module 2

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D

1	A	B	C	D
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4	A	B	C	D
5	A	B	C	D
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8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D

Answer sheet

SAT Practice Test 2, Math

Fill in the circle for each answer, or write your student-produced answers in the boxes. Then mark yourself with the answer key.

Module 1

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
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13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
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25				
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27				

Module 2

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
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14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D
20	A	B	C	D
21				
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24				
25				
26				
27				

Answer key

SAT Practice Test 2, Reading and Writing

Module 1

NO.	ANSWER	SKILL
1	C	Words in context
2	A	Words in context
3	D	Words in context
4	B	Words in context
5	A	Words in context
6	B	Words in context
7	C	Text structure and purpose
8	D	Text structure and purpose
9	B	Text structure and purpose
10	C	Central ideas and details
11	D	Central ideas and details
12	A	Command of evidence: quantitative
13	A	Command of evidence: quantitative
14	D	Command of evidence: textual
15	C	Command of evidence: textual
16	B	Command of evidence: textual
17	A	Inferences
18	B	Inferences
19	D	Inferences
20	C	Form, structure, and sense
21	C	Boundaries
22	A	Form, structure, and sense
23	B	Boundaries
24	D	Form, structure, and sense
25	D	Boundaries
26	B	Form, structure, and sense
27	C	Boundaries
28	A	Transitions
29	A	Transitions
30	B	Transitions
31	D	Rhetorical synthesis
32	C	Rhetorical synthesis
33	A	Rhetorical synthesis

Module 2

NO.	ANSWER	SKILL
1	A	Words in context
2	D	Words in context
3	B	Words in context
4	C	Words in context
5	B	Words in context
6	A	Words in context
7	C	Text structure and purpose
8	D	Text structure and purpose
9	C	Text structure and purpose
10	D	Cross-text connections
11	B	Central ideas and details
12	A	Central ideas and details
13	A	Command of evidence: quantitative
14	C	Command of evidence: quantitative
15	D	Command of evidence: textual
16	B	Command of evidence: textual
17	A	Command of evidence: textual
18	C	Inferences
19	B	Inferences
20	D	Inferences
21	B	Form, structure, and sense
22	C	Boundaries
23	A	Form, structure, and sense
24	D	Boundaries
25	B	Boundaries
26	D	Form, structure, and sense
27	C	Form, structure, and sense
28	A	Transitions
29	A	Transitions
30	C	Transitions
31	D	Rhetorical synthesis
32	B	Rhetorical synthesis
33	A	Rhetorical synthesis

Answer key

SAT Practice Test 2, Math

Module 1

NO.	ANSWER	SKILL
1	**B**	Percentages
2	**D**	Data analysis
3	**A**	Linear equations
4	**C**	Systems of equations
5	**C**	Functions
6	**A**	Nonlinear expressions
7	**D**	Ratios and rates
8	**B**	Triangles
9	**D**	Circles
10	**A**	Quadratics
11	**C**	Probability
12	**B**	Statistics
13	**C**	Scatterplots
14	**B**	Inequalities
15	**D**	Functions
16	**A**	Exponential functions
17	**C**	Linear functions
18	**B**	Percent change
19	**A**	Volume
20	**D**	Right triangles
21	**16**	Systems of equations
22	**4**	Quadratics
23	**15**	Percent change
24	**11**	Linear equations
25	**9**	Functions
26	**8**	Right triangles
27	**56**	Area

Module 2

NO.	ANSWER	SKILL
1	**A**	Exponential functions
2	**C**	Quadratics
3	**D**	Percentages
4	**B**	Volume
5	**C**	Functions
6	**A**	Statistics
7	**D**	Probability
8	**B**	Percentages
9	**C**	Equivalent expressions
10	**A**	Percentages
11	**D**	Systems of equations
12	**B**	Surface area
13	**A**	Circles
14	**B**	Equivalent expressions
15	**D**	Trigonometry
16	**C**	Exponential equations
17	**A**	Exponential functions
18	**B**	Linear functions in context
19	**D**	Absolute value
20	**C**	Ratios and rates
21	**10**	Linear equations
22	**8**	Circles in the plane
23	**8**	Quadratics
24	**154**	Quadratics
25	**15**	Percentages
26	**14**	Quadratics
27	**-3**	Systems with nonlinear equations

Score yourself. Count the questions you got right in each module.

Reading and Writing, Module 1: ___ of 33

Module 2: ___ of 33

Math, Module 1: ___ of 27

Module 2: ___ of 27

Answer explanations

SAT Practice Test 2, Reading and Writing, Module 1

Questions 1 to 17

1 **Choice C** Words in context

Restricting the movement of firewood is a way of keeping the beetle from spreading, so limit fits. Encourage and celebrate run against the officials' aim, and document does not describe a restriction.

2 **Choice A** Words in context

The colon explains that the menu has only five dishes, so limited is the logical word. Sprawling is the opposite, and the passage says nothing about indifference or age.

3 **Choice D** Words in context

She repeats herself to avoid disappointing buyers, which shows a reluctance to take risks that complacent captures. Curious would lead her to experiment, and nothing suggests generosity or impatience.

4 **Choice B** Words in context

Dense shade that blocks seedlings suggests very large, dominant trees, and imposing conveys that. Delicate and unremarkable would not cast such shade.

5 **Choice A** Words in context

Rather than signals a contrast with keeping the walls and beams, so the developers chose not to demolish the building. The other words do not contrast with conversion.

6 **Choice B** Words in context

The speech had no specific proposals and left listeners unsure of her intentions, so vague is the logical word. Precise is the opposite, and the speech was long, not brief.

7 **Choice C** Text structure and purpose

The underlined sentence says what the angle and duration of the dance mean, which explains how the dance carries information. It does not question bee communication or compare other insects.

8 **Choice D** Text structure and purpose

Lucía sets everything back where it was and cannot turn the calendar page, which shows her reluctance to disturb her aunt's life. She is careful, not careless, and the text does not explain the tidiness or suggest haste.

9 **Choice B** Text structure and purpose

The first stanza describes a river unaware of human things, and the second turns to people hoping that time will slow. No history, remedy or comparison of rivers appears.

10 **Choice C** Central ideas and details

The text says the harvests are modest and that the real yield is social. It denies that the gardens could feed a neighbourhood and says nothing about difficulty or about most residents.

11 **Choice D** Central ideas and details

The text says the viaduct's failure is what she is remembered for, but that a contractor swapped her steel for a cheaper alloy. Nothing is said about destroyed notebooks, dismissing vibration, or being first to use steel.

12 **Choice A** Command of evidence: quantitative

The claim compares the share preferring loose fruit in the older and younger groups, which the first choice does. The next uses the pre-packed share, and the last mixes the two measures, so neither compares the same preference.

13 **Choice A** Command of evidence: quantitative

A recovery after closure means citing the low in 2020 and the later rise, and the first choice does this with figures. The others mix up the years or leave out the low point.

14 **Choice D** Command of evidence: textual

The sparseness of the later music is shown by the work asking the orchestra to play almost nothing. The first quotation describes the early dense style, and the others are either not about sparseness or describe its effect.

15 **Choice C** Command of evidence: textual

Pages folded and stitched around the cover show the fragment was used to bind the ledger. The language, the contents of the ledger and other holdings do not address how the fragment was used.

16 **Choice B** Command of evidence: textual

Proponents say tides can be predicted far in advance, making the power easy to plan around. Cost and wildlife concerns are raised by critics, and the text does not compare the strength of tides and winds.

17 **Choice A** Inferences

Seeds that germinate in response to smoke are adapted to take advantage of conditions after fires. The text does not say they cannot grow elsewhere, and it contradicts the other choices.

Answer explanations

SAT Practice Test 2, Reading and Writing, Module 1

Questions 18 to 33

18 Choice B Inferences

The researcher notes the cause might run the other way, so the survey alone cannot show that reading builds vocabulary. It does not show harm, claim that weak readers never read, or restrict the findings to one university.

19 Choice D Inferences

Rueda calls the year of study without performance his most formative, which suggests he values study over performance. He does not say that young people should not perform, nor criticise travel or scores.

20 Choice C Form, structure, and sense

The adverbial phrase next week requires the future, and the plural subject students takes will present. The other forms are singular or past.

21 Choice C Boundaries

The three items form a parallel list of plural nouns, so textiles is correct. The other choices break the parallel structure.

22 Choice A Form, structure, and sense

The opening phrase must modify the person who did the studying, so Dr. Kim must be the subject that follows the comma. The other choices attach the phrase to findings, the meeting or a pronoun, creating a dangling modifier.

23 Choice B Boundaries

The relative clause which took nearly four years to build is nonessential and is closed by a comma before the main verb. A semicolon, colon or full stop would leave the main clause without a subject.

24 Choice D Form, structure, and sense

The subject each is singular, so the verb is receives. The other choices are plural or ungrammatical.

25 Choice D Boundaries

The semicolon joins two independent clauses, and the second is a result of the first, so as a result is correct.

26 Choice B Form, structure, and sense

The second clause gives the reason the discovery was important, so because is the logical conjunction.

27 Choice C Boundaries

In this inverted sentence the subject, mask and cloak, comes after the verb and is plural, so the verb must be were.

28 Choice A Transitions

The botanical classification contrasts with the culinary one, so however is the right transition.

29 Choice A Transitions

The fall in commuting times confirms what was expected, so Indeed is the logical transition.

30 Choice B Transitions

The sentence corrects the earlier belief with a fact, so In fact is the best transition. Likewise suggests similarity, and the others do not fit.

31 Choice D Rhetorical synthesis

To explain what the building is for, the choice must state its function, and only the first mentions tracking variable stars. The others give its history, visiting hours or an installation date.

32 Choice C Rhetorical synthesis

A comparison needs details about both rivers on the same points, which the first choice gives. The others describe one river or the shared sea.

33 Choice A Rhetorical synthesis

Age is conveyed by the date of around 1190, which only the first choice gives. The others describe the codex's contents, material or restoration.

Answer explanations

SAT Practice Test 2, Reading and Writing, Module 2

Questions 1 to 17

1 **Choice A** Words in context

She said enough to be friendly but nothing binding, so her remarks were carefully measured. Calibrated fits. Impulsive and defiant contradict the care described, and garbled would mean confused.

2 **Choice D** Words in context

The sentence describes a disease once confined to the tropics, so scourge, meaning a source of affliction, is the fitting noun. Haven and remedy suggest benefit, and relic implies something from the past.

3 **Choice B** Words in context

Wording that lets both sides claim a win must be capable of several readings, which elastic suggests. Rigid and emphatic would leave little room for dispute.

4 **Choice C** Words in context

Far from signals that she was the opposite of resistant, and she actively sought criticism. Impervious means unable to be affected, which fits the contrast. Vulnerable and sensitive would not contrast with seeking criticism.

5 **Choice B** Words in context

The colleagues preferred established procedures, so the methods must have been unconventional to them. Unorthodox fits. Conventional is the opposite, and the methods were later adopted, so they were not negligent.

6 **Choice A** Words in context

Although signals a contrast with the survey's finding that his descriptions were accurate, so the journals must have been dismissed as exaggerated. Verified and celebrated would not contrast with the survey, and transcribed does not fit with as exaggerated.

7 **Choice C** Text structure and purpose

The underlined sentence describes the slow accumulation of discrepancies, which shows that the revolution was gradual, contrary to the popular image of one decisive moment. It does not describe a single experiment, say the old model had no defenders, or introduce a theory to be evaluated.

8 **Choice D** Text structure and purpose

The sentence lists real hazards the speeches ignored, which supports Corliss's impression that the speakers lack knowledge of the sea. It does not mention a speech by him, claim the vessel is more dangerous, or say the harbour is now safe.

9 **Choice C** Text structure and purpose

The first sentence reports what the speaker was told, and the rest describes a house of language that has been filled with outside words. No house is praised, advice followed or languages ranked.

10 **Choice D** Cross-text connections

Text 2 notes the new styles began before photography spread, so the historian would say Text 1 mistakes cause for coincidence or acceleration. The historian says the camera made visible an existing interest, so does not deny all influence.

11 **Choice B** Central ideas and details

The text reports membraneless droplets that act as compartments, which means cells need less physical structure than thought. It says the picture is more crowded, not less, and oil is only an analogy.

12 **Choice A** Central ideas and details

The text argues that GDP counts all spending, including repair of damage, so it should not be read as welfare. It does not say GDP falls with repairs, or that economists have abandoned it.

13 **Choice A** Command of evidence: quantitative

At week 12 the graph shows 92 cm for the higher amount and 77 cm for the lower, a difference of 15 cm. The other choices change the size of the gap, swap the groups, or move the data to the wrong week.

14 **Choice C** Command of evidence: quantitative

Reliability concerns how little the time varies, so the correct choice compares the differences between rush hour and other times. One choice reports only rush hour times, one swaps the figures for the bus and car, and one reports the shortest time rather than the variation.

15 **Choice D** Command of evidence: textual

The claim is that a mistaken theory could still lead to effective remedies, and only the quotation saying the measures worked but not for the believed reason states this directly. The others describe the theory, the measures or one mechanism without making the point.

Answer explanations

SAT Practice Test 2, Reading and Writing, Module 2

Questions 18 to 33

18 Choice C Inferences

Narrow, winding streets inside walls contrasted with wider ones built once the threat lessened, which suggests the earlier layout reflected defensive needs. Speed and shared architects are not suggested, and the walls are connected to the layout.

19 Choice B Inferences

Dense cordgrass raised the surface in step with sea level while thin cover let the marsh drown, so restoring grass should help marshes survive. The text gives no support for sinking or for calm seas making grass unnecessary.

20 Choice D Inferences

Identical copies with the same page numbers let scholars point readers to the same words, so they could discuss and check claims against a shared text. The text does not say scholars avoided earlier work or relied on handwriting.

21 Choice B Form, structure, and sense

The relative pronoun is the subject of had spent and refers to people, so who is correct. Whom is the wrong case, and which and that they are not suitable.

22 Choice C Boundaries

The subject is the singular number, not the plural species, so has agrees. A number of would take a plural verb, but the number of takes a singular one.

23 Choice A Form, structure, and sense

The opening phrase compares Pryce with her contemporaries, so the subject that follows must be Pryce herself. The other choices make the countryside or her writing the thing being compared with people.

24 Choice D Boundaries

The items in this list already contain commas, so semicolons are used to separate them. Only the first choice separates the second and third items with a semicolon and the conjunction and.

25 Choice B Boundaries

The subject is the singular fraction, so the verb is is. The other choices do not agree or are not finite verbs.

26 Choice D Form, structure, and sense

After the negative adverb rarely, the auxiliary comes before the subject, so does an observer encounter is correct. The other choices fail to invert or fail to agree.

27 Choice C Form, structure, and sense

The streets belong to the city, which requires the singular possessive city's. Citys is not a word, and the plural forms do not fit one city.

28 Choice A Transitions

The sentence introduces newer findings that qualify the long-standing recommendation, so More recently, is the correct transition.

29 Choice A Transitions

The poets' embrace of railways contrasts with the general suspicion, so Nevertheless is the logical transition.

30 Choice C Transitions

The short lifespan is a limit that contrasts with the animal's intelligence, so Yet is the correct transition.

31 Choice D Rhetorical synthesis

The goal is to show how a discovery changed what is known, so the choice must contrast the earlier count with the newer one, which the first does. The others give dates, a detail of the new figures, or a method without the change.

32 Choice B Rhetorical synthesis

The audience wants the key result and its limit, and only the first choice gives both the success rate and the false alarms. The others give the method, a call for trials, or the success rate alone.

33 Choice A Rhetorical synthesis

A contrast needs both approaches with their features, which the first choice supplies. The others name the methods without features, describe only one method, or give a single reason.

Answer explanations

SAT Practice Test 2, Math, Module 1

Questions 1 to 14

1 **Choice B** Percentages

20% of 500 is 0.2 times 500, which is 100.

2 **Choice D** Data analysis

Team 5 has 25 and Team 1 has 15, and 25 minus 15 is 10.

3 **Choice A** Linear equations

Subtract 2x from both sides: 7x + 2 = 72. Then 7x = 70, so x = 10.

4 **Choice C** Systems of equations

Adding the equations gives 2x = 28, so x = 14.

5 **Choice C** Functions

Each step of 1 in x adds 6 to f(x), so the slope is 6. Then f(1) = 14 gives 6 + 8, so the intercept is 8.

6 **Choice A** Nonlinear expressions

Multiply the coefficients, 5 times 5 is 25, and add the exponents, 5 plus 6 is 11.

7 **Choice D** Ratios and rates

6 cups of sugar is 3 times 2 cups, so the flour is 3 times 7 cups, which is 21.

8 **Choice B** Triangles

The angles of a triangle add to 180°, so x + 2x + 6x = 9x = 180 and x = 20.

9 **Choice D** Circles

The circumference $2\pi r = 22\pi$ gives r = 11. The area is $\pi r^2 = 121\pi$.

10 **Choice A** Quadratics

The equation factors as (x − 8)(x + 4) = 0, so x = 8 or x = −4.

11 **Choice C** Probability

45 students travel that way and 18 of them are in Grade 10, so the probability is 18/45, which simplifies to 2/5. The fraction with 138 on the bottom uses every student.

12 **Choice B** Statistics

The first 5 scores total 345, and adding 81 gives 426. Divided by 6, the mean is 71.

13 **Choice C** Scatterplots

Substitute x = 6: 6(6) + 44 = 36 + 44 = 80.

14 **Choice B** Inequalities

The total is the 32 already joined plus 10 each week, and it must be at least 150, so 32 + 10w ≥ 150.

Answer explanations

SAT Practice Test 2, Math, Module 1

Questions 15 to 27

15 **Choice D** Functions

f(4) = 4(16) − 12 + 6 = 64 − 12 + 6 = 58.

16 **Choice A** Exponential functions

Tripling every 6 hours means the number of triplings is t/6, so the population is 800 times 3 to the power t/6.

17 **Choice C** Linear functions

The slope is (28 − 13)/(6 − 3) = 5. Using (3, 13), 13 = 5(3) + b gives b = −2.

18 **Choice B** Percent change

The increase is 1,500, and dividing by the original 6,000 gives 0.25, which is 25%. Dividing by the new value would give a smaller percent.

19 **Choice A** Volume

The volume is $\pi r^2 h = \pi(36)(8) = 288\pi$.

20 **Choice D** Right triangles

$5^2 + 12^2 = 169$, so the hypotenuse is $\sqrt{169} = 13$.

21 **16** Systems of equations

Adding x + y = 22 and x − y = 10 gives 2x = 32, so the larger number is 16.

22 **4** Quadratics

The equation factors as (x − 4)(x + 3) = 0, so the solutions are 4 and −3. The positive one is 4.

23 **15** Percent change

The increase is 3 dollars, and 3 divided by the original 20 is 0.15, which is 15 percent.

24 **11** Linear equations

Expanding gives 4x − 8 = 8x − 52. Collecting terms gives −4x = −44, so x = 11.

25 **9** Functions

8a − 6 = 66 gives 8a = 72, so a = 9.

26 **8** Right triangles

By the Pythagorean theorem, the other leg is $\sqrt{(100 - 36)} = \sqrt{64} = 8$.

27 **56** Area

The area of a triangle is half the base times the height: (1/2)(16)(7) = 56.

Answer explanations

SAT Practice Test 2, Math, Module 2

Questions 1 to 14

1 **Choice A** Exponential functions

Tripling every 2 hours means the number of triplings is t/2, so the population is 500 times 3 to the power t/2.

2 **Choice C** Quadratics

The equation factors as $(x - 8)(x + 1) = 0$, so $x = 8$ or $x = -1$.

3 **Choice D** Percentages

80% of 300 is 0.8 times 300, which is 240.

4 **Choice B** Volume

The volume is $\pi r^2 h = \pi(9)(7) = 63\pi$.

5 **Choice C** Functions

$f(4) = 3(16) - 24 + 5 = 48 - 24 + 5 = 29$.

6 **Choice A** Statistics

The first 6 scores total 480, and adding 94 gives 574. Divided by 7, the mean is 82.

7 **Choice D** Probability

37 students travel that way and 25 of them are in Grade 10, so the probability is 25/37. The fraction with 99 on the bottom uses every student.

8 **Choice B** Percentages

Start at 100. After the 25% rise the price is 125, and 40% off 125 leaves 75. That is 25% lower than the original.

9 **Choice C** Equivalent expressions

$(4x + 3)^2 = 16x^2 + 24x + 9$, and $(3x - 4)^2 = 9x^2 - 24x + 16$. Subtracting the second from the first gives $7x^2 + 48x - 7$.

10 **Choice A** Percentages

150 is 90 more than 60, and 90 / 60 = 1.5, so p = 150.

11 **Choice D** Systems of equations

A system has no solution when the lines are parallel but not identical. The second equation is 3 times the left side of the first with a different constant, so a must equal 15/3 = 5.

12 **Choice B** Surface area

Each cube has 6 faces, and gluing two faces together hides 2 of the 12, leaving 10 faces of area s^2. So $10s^2 = 640$, $s^2 = 64$, and $s = 8$.

13 **Choice A** Circles

Arc length is the radius times the central angle in radians: $9 \times \pi/6 = 9/6\pi$.

14 **Choice B** Equivalent expressions

The numerator factors as $(x - 7)(x - 5)$. Cancelling $x - 7$ leaves $x - 5$.

Answer explanations

SAT Practice Test 2, Math, Module 2

Questions 15 to 27

15 **Choice D** Trigonometry

A right triangle with opposite side 3 and hypotenuse 5 has adjacent side 4, so tan A = opposite/adjacent = 3/4.

16 **Choice C** Exponential equations

Write 4 as 2^2, so the right side is 2^(2x − 6). Equal bases give x + 4 = 2x − 6, so x = 10.

17 **Choice A** Exponential functions

A year is 12 months, so the value is multiplied by 0.9^(12/6) = 0.9^2 = 0.81 each year. That is a decrease of 1 − 0.81, or 19%.

18 **Choice B** Linear functions in context

The coefficient of x is the change in cost for each extra hour. The constant 45 is the fixed fee, and the 1 hour cost would be 48.5.

19 **Choice D** Absolute value

The inequality means x is within 3 of 6, so 3 ≤ x ≤ 9. That range has 7 integers, counting both ends.

20 **Choice C** Ratios and rates

The pumps fill 1/12 and 1/6 of the tank each hour, together 1/4 of it, so the time is 4 hours.

21 **10** Linear equations

Expanding gives 4x − 12 = 8x − 52. Collecting terms gives −4x = −40, so x = 10.

22 **8** Circles in the plane

Complete the square: $(x + 1)^2 + (y + 5)^2 = 38 + 1 + 25 = 64$. The radius is √64 = 8.

23 **8** Quadratics

For $ax^2 + bx + c = 0$ the solutions add to −b/a = −(−16)/2 = 8.

24 **154** Quadratics

The maximum is at t = −b/(2a) = 48/8 = 6. Then h(6) = −4(36) + 288 + 10 = 154.

25 **15** Percentages

Let L be the litres of the 20% solution. Then 0.2L + 0.7(10) = 0.4(L + 10). Solving gives L = 15.

26 **14** Quadratics

Exactly one real solution means the discriminant is zero: $b^2 − 4(49) = 0$, so $b^2 = 196$ and b = 14 (taking the positive value).

27 **-3** Systems with nonlinear equations

Setting the expressions equal gives $x^2 − 6x + (6 − k) = 0$. One solution means the discriminant is 0: 36 − 4(6 − k) = 0, so k = −3.